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Transport device and recording device

Abstract

A transport device includes a transporting belt having an endless shape and configured to transport a medium, a first roller including a first circumferential surface brought into contact with the transporting belt, the first roller being a roller around which the transporting belt is wound, a second roller including a second circumferential surface brought into contact with the transporting belt, the second roller being a roller around which the transporting belt is wound, a pressing section including a bearing unit including a contact portion brought into contact with the first circumferential surface, the pressing section being configured to press the first circumferential surface on an inner side of the transporting belt, and a support portion disposed on an inner side of the transporting belt and configured to support the pressing section.

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Patent No.	Application Date	Country	CPC
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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-039993, filed on Mar. 15, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a transport device and a recording device.

2. Related Art

(3) There are known a transport device configured to transport a printing medium, and a printing apparatus that includes the transport device. A transport device described in JP-A-2021-147181 includes a belt transport roller, a belt-rotated roller, and a transporting belt. In addition, the transport device includes a tension adjustment unit. The tension adjustment unit biases a base portion configured to support the belt-rotated roller. By biasing the base portion, the tension adjustment unit adjusts the belt tension of the transporting belt. The base portion supports a shaft end portion of the belt-rotated roller.

(4) In a case of the transport device that transports the printing medium using the transporting belt, a difference in peripheral speeds of the transporting belt may vary depending on locations within the transporting belt.

SUMMARY

(5) A transport device according to the present disclosure includes a transporting belt having an endless shape and configured to transport a medium, a first roller including a first circumferential surface brought into contact with the transporting belt, the first roller being a roller around which

the transporting belt is wound, a second roller including a second circumferential surface brought into contact with the transporting belt, the second roller being a roller around which the transporting belt is wound, a pressing section including a bearing unit including a contact portion brought into contact with the first circumferential surface, the pressing section being configured to press the first circumferential surface on an inner side of the transporting belt, and a support portion disposed on an inner side of the transporting belt and configured to support the pressing section.

(6) A recording device according to the present disclosure includes a transporting belt having an endless shape and configured to transport a medium, a first roller including a first circumferential surface brought into contact with the transporting belt, the first roller being a roller around which the transporting belt is wound, a second roller including a second circumferential surface brought into contact with the transporting belt, the second roller being a roller around which the transporting belt is wound, a pressing section including a bearing unit including a contact portion brought into contact with the first circumferential surface, the pressing section being configured to press the first circumferential surface on an inner side of the transporting belt, a support portion disposed on an inner side of the transporting belt and configured to support the pressing section, and a recording unit configured to perform recording on the medium transported by the transporting belt.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a schematic configuration of a printer.
- (2) FIG. 2 is a diagram illustrating a schematic configuration of a transport unit as viewed from a +Z direction.
- (3) FIG. 3 is a diagram illustrating a schematic configuration of the transport unit as viewed from a +Y direction.
- (4) FIG. 4 is a diagram illustrating a schematic configuration of the transport unit as viewed from the +Y direction.
- (5) FIG. 5 is a diagram illustrating a schematic configuration of the transport unit as viewed from the +Z direction.
- (6) FIG. 6 is a diagram illustrating a schematic configuration of the transport unit as viewed from the +Z direction.
- (7) FIG. 7 is a diagram illustrating a schematic configuration of the transport unit as viewed from the +Y direction.
- (8) FIG. 8 is a diagram illustrating a schematic configuration of the transport unit including a pressing-force adjusting mechanism as viewed from the +Y direction.
- (9) FIG. 9 is a diagram illustrating a schematic configuration of the transport unit including the pressing-force adjusting mechanism as viewed from the +Y direction.
- (10) FIG. 10 is a diagram illustrating a schematic configuration of the transport unit including the pressing-force adjusting mechanism as viewed from the +Y direction.
- (11) FIG. 11 is a diagram illustrating a schematic configuration of the transport unit including the pressing-force adjusting mechanism as viewed from the +Y direction.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (12) FIG. 1 illustrates a schematic configuration of a printer 1. The printer 1 performs printing on a printing medium M. As one example, the printer 1 is an inkjet printer configured to discharge ink on the printing medium M to perform printing on the printing medium M. The printer 1 corresponds to one example of a recording device. The printing medium M corresponds to one example of a medium.
- (13) The plurality of figures including FIG. 1 illustrate a XYZ coordinate system. The X-axis, the

Y-axis, and the Z-axis are perpendicular to each other. The Z-axis is an axis perpendicular to the installation surface of the printer **1** that is not illustrated. The X-axis and the Y-axis are parallel to the installation surface. The X-axis is an axis extending along a direction in which the upper surface of a medium transporting belt **10**, which will be described later, moves. The Y-axis is an axis intersecting, at a right angle, the direction in which the upper surface of the medium transporting belt **10** moves. A direction extending along the Z-axis from the installation surface toward the upward direction is referred to as the +Z direction. A direction extending along the Z-axis from the installation surface toward the downward direction is referred to as the -Z direction. The direction in which the upper surface of the medium transporting belt **10** moves is referred to as a +X direction. The reverse direction to the direction in which the upper surface of the medium transporting belt **10** moves is referred to as a -X direction. A direction from the rear direction of the printer **1** illustrated in FIG. **1** toward the front direction is referred to as a +Y direction. A direction from the front direction of the printer **1** illustrated in FIG. **1** toward the rear direction is referred to as a -Y direction.

(14) The printer **1** includes a feeding unit **2**, a transport unit **3**, a printing unit **4**, a cleaning unit **5**, and a winding unit **6**. The printer **1** illustrated in FIG. **1** performs printing on the printing medium **M** wound around the feeding unit **2**. The printer **1** is an inkjet printer configured to discharge ink on the printing medium **M** to perform printing.

(15) The feeding unit **2** feeds the printing medium **M** from a medium roll **R1**. The medium roll **R1** is a roll obtained by rolling the printing medium **M** into a roll form. The feeding unit **2** includes a roll rotary shaft **21**, a rotary-shaft driving mechanism, which is not illustrated in the drawing, and a guide roller **23**.

(16) The roll rotary shaft **21** rotatably supports the medium roll **R1**. The medium roll **R1** is installed at the roll rotary shaft **21** by a user. The roll rotary shaft **21** rotates in a rotational direction **C** to cause the printing medium **M** to be fed from the medium roll **R1**. The fed printing medium **M** is transported in a transport direction **A**.

(17) The rotary-shaft driving mechanism causes the roll rotary shaft **21** to rotate in the rotational direction **C**. The rotary-shaft driving mechanism causes the roll rotary shaft **21** to rotate, thereby feeding the printing medium **M** from the medium roll **R1**. The rotary-shaft driving mechanism may adjust the tension of the printing medium **M** that is being fed.

(18) The guide roller **23** guides the printing medium **M**. The guide roller **23** transports, to the transport unit **3**, the printing medium **M** fed from the medium roll **R1**. The guide roller **23** may adjust the tension of the printing medium **M**. With a tension adjusting mechanism that is not illustrated in the drawing, the guide roller **23** moves in the +Z direction or the -Z direction. As the guide roller **23** moves in the +Z direction or the -Z direction, the guide roller **23** adjusts the tension of the printing medium **M**.

(19) The transport unit **3** transports the printing medium **M** to a position that is opposed to the printing unit **4**. The transport unit **3** includes a driving roller **8**, a driving mechanism that is not illustrated in the drawing, a driven roller **9**, the medium transporting belt **10**, a pressure applying roller **12**, a support frame **14**, and a pressing unit **16**. The transport unit **3** corresponds to one example of a transport device.

(20) The driving roller **8** rotationally moves the medium transporting belt **10**. The driving roller **8** is disposed at the +X direction of the printing unit **4**. The driving roller **8** includes a driving-roller circumferential surface **8a**. The driving-roller circumferential surface **8a** is brought into contact with the medium transporting belt **10**. The driving-roller circumferential surface **8a** transmits driving force of the driving roller **8** to the medium transporting belt **10**. The driving roller **8** corresponds to one example of a first roller or a second roller. The driving-roller circumferential surface **8a** corresponds to one example of a first circumferential surface or a second circumferential surface.

(21) The driving roller **8** illustrated in FIG. **1** is a straight roller configured to have an outer

diameter constant within tolerance limits. It is preferable that the driving roller **8** is a crown roller configured such that the outer diameter of the central portion, in the rotary shaft direction, of the driving roller **8** is greater than the outer diameter of each end portion, in the rotary shaft direction, of the driving roller **8**. For the driving roller **8** having a length of not less than 1000 mm, it is preferable that a difference between the outer diameter of the central portion, in the rotary shaft direction, of the crown roller and the outer diameter of each end portion in the rotary shaft direction falls in a range of not less than 5 μm and not more than 200 μm .

(22) It is preferable that the driving roller **8** is a crown roller.

(23) With the driving roller **8** being configured as a crown roller, it is possible to reduce deformation, in the $-X$ direction, of the central portion, in the rotary shaft direction, of the driving roller **8** due to the tension of the medium transporting belt **10**.

(24) The driving mechanism causes the driving roller **8** to rotationally drive. The driving mechanism generates rotational driving force. The driving mechanism transmits the generated rotational driving force to the driving roller **8**. By transmitting the rotational driving force, the driving mechanism causes the driving roller **8** to rotate. The driving mechanism transmits the rotational driving force from the shaft end portion of the driving roller **8**.

(25) The driven roller **9** supports the medium transporting belt **10** so as to be able to rotationally move. The driven roller **9** is disposed at the $-X$ direction of the printing unit **4**. The driven roller **9** is rotatably supported by a main body frame or the like that is not illustrated in the drawing. The driven roller **9** includes a driven-roller circumferential surface **9a**. The driven-roller circumferential surface **9a** is brought into contact with the medium transporting belt **10**. When the medium transporting belt **10** rotationally moves, the driven-roller circumferential surface **9a** followingly rotates. The driven roller **9** corresponds to one example of a first roller or a second roller. The driven-roller circumferential surface **9a** corresponds to one example of a first circumferential surface or a second circumferential surface.

(26) The driven roller **9** illustrated in FIG. **1** is a straight roller configured to have an outer diameter constant within tolerance limits. The driven roller **9** may be a crown roller configured such that the outer diameter of the central portion, in the rotary shaft direction, of the driven roller **9** is greater than the outer diameter of each end portion, in the rotary shaft direction, of the driven roller **9**. For the driven roller **9** having a length of not less than 1000 mm, it is preferable that a difference between the outer diameter of the central portion, in the rotary shaft direction, of the crown roller and the outer diameter of each end portion falls in a range of not less than 5 μm and not more than 200 μm .

(27) The medium transporting belt **10** transports the printing medium **M**. The medium transporting belt **10** is a belt having an endless shape. The medium transporting belt **10** is an adhesive belt having an outer peripheral surface provided with an adhesive layer **F**. The adhesive layer **F** includes an adhesive. The adhesive is a hot-melt based adhesive containing thermoplastic elastomer **SIS** as a primary component. The “**SIS**” stands for styrene-isoprene block copolymers. For example, the adhesive includes “**Polixresin**”, “**Newdine**”, and “**Aquadine**” series manufactured by Yokohama Polymer Co., Ltd, “**MC polymer series**” manufactured by Murayama Chemical Research, “**UNIKENSOL RV-30 (for screen printing)**” manufactured by Union Chemical Industry Co., Ltd, “**Plaster EH**” manufactured by Shin-Nakamura Chemical Co, Ltd, “**ATRASOL GP1 (ATR code: ATR1717)**” manufactured by ATR CHEMICALS SA, or the like. With the adhesive layer **F**, the medium transporting belt **10** causes the printing medium **M** to adhere to the outer peripheral surface thereof. The printing medium **M** fed from the feeding unit **2** is mounted on the medium transporting belt **10**. The medium transporting belt **10** transports, in the $+X$ direction, the printing medium **M** mounted thereon. The medium transporting belt **10** illustrated in FIG. **1** is extended in a tension manner by the driving roller **8** and the driven roller **9**. The medium transporting belt **10** corresponds to one example of a transporting belt.

(28) Although the medium transporting belt **10** illustrated in FIG. **1** is extended in a tension manner

by the driving roller **8** and the driven roller **9**, the configuration thereof is not limited to this. The medium transporting belt **10** may be extended in a tension manner by three or more rollers including the driving roller **8** and the driven roller **9**.

(29) The medium transporting belt **10** illustrated in FIG. **1** is an adhesive belt but is not limited to this. The medium transporting belt **10** may be an electrostatic adhering belt configured to electrostatically draw the printing medium M. There is no limitation applied to the mode of the medium transporting belt **10**, provided that the medium transporting belt **10** has a function of holding the printing medium M and transporting the held printing medium M. It is preferable that the medium transporting belt **10** is an adhesive belt. The adhesive belt is able to hold various types of printing medium M such as fabric, and transport the held printing medium M.

(30) The pressure applying roller **12** applies pressure to the printing medium M against the outer peripheral surface of the medium transporting belt **10**. The pressure applying roller **12** applies pressure to the printing medium M guided by the guide roller **23**, so as to be pressed onto the medium transporting belt **10**. The pressure applying roller **12** causes the printing medium M to adhere to the outer peripheral surface of the medium transporting belt **10**. With the pressure applied by the pressure applying roller **12**, the printing medium M is caused to stick to the outer peripheral surface of the medium transporting belt **10**. The printing medium M caused to stick is transported by the medium transporting belt **10** to a position that is opposed to the printing unit **4**.

(31) The support frame **14** is configured to support the pressing unit **16**. The support frame **14** is disposed on an inner side of the medium transporting belt **10**. The support frame **14** may be supported by a main body frame that is not illustrated in the drawing. The main body frame is disposed at the +Y direction of the medium transporting belt **10** and also at the -Y direction. The support frame **14** may constitute a platen that is opposed to the printing unit **4** with the medium transporting belt **10** being interposed therebetween. The support frame **14** may be in contact with the inner peripheral surface of the medium transporting belt **10** or may not be in contact with it. The support frame **14** corresponds to one example of a support portion.

(32) The pressing unit **16** presses the driving roller **8** or a portion of the driven roller **9**. The pressing unit **16** illustrated in FIG. **1** is brought into contact with the driving-roller circumferential surface **8a** of the driving roller **8**. The pressing unit **16** is brought into contact with the driving-roller circumferential surface **8a** and presses the driving roller **8**. The pressing unit **16** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The structure of the pressing unit **16** will be described later. The pressing unit **16** corresponds to one example of a pressing section.

(33) The printing unit **4** performs printing on the printing medium M. The printing unit **4** includes a printing head **41**, a carriage **43**, and a carriage driving mechanism that is not illustrated in the drawing. The printing unit **4** illustrated in FIG. **1** discharges ink on the printing medium M to perform printing. The printing unit **4** may be a laser type or the like configured to cause toner to be attached on the printing medium M to perform printing. The printing unit **4** corresponds to one example of a recording unit. Printing corresponds to one example of recording.

(34) The printing head **41** discharges ink onto the printing medium M. The printing head **41** discharges ink onto the printing medium M to perform printing. The printing head **41** is an inkjet head. The printing head **41** discharges one or more types of ink.

(35) The carriage **43** supports the printing head **41**, and moves it along a drive axis B. The drive axis B is an imaginary axis extending along the Y-axis. The carriage **43** moves along the drive axis B and in the +Y direction and the -Y direction. The carriage **43** moves the supported printing head **41** along the drive axis B and in the +Y direction and the -Y direction. When the carriage **43** moves in the +Y direction or the -Y direction, the printing head **41** discharges ink onto the printing medium M.

(36) The carriage driving mechanism causes the carriage **43** to move. The carriage driving mechanism generates driving force that causes the carriage **43** to move. The carriage driving

mechanism transmits the generated driving force to the carriage **43**. With the driving force transmitted to the carriage **43**, the carriage **43** moves along the drive axis B.

(37) The printing unit **4** illustrated in FIG. **1** includes the carriage **43**. However, the configuration thereof is not limited to this configuration. The printing unit **4** may not include the carriage **43**. As one example, when the printing head **41** is a line inkjet head including a nozzle row having a width equal to or larger than the width of the printing medium M along the Y-axis, the carriage **43** may not be used.

(38) The cleaning unit **5** cleans the medium transporting belt **10**. The cleaning unit **5** is provided at the $-Z$ direction of the medium transporting belt **10**. The cleaning unit **5** cleans the medium transporting belt **10** that moves from the driving roller **8** to the driven roller **9**. The cleaning unit **5** includes a cleaning brush **51**, a cleaning-brush driving mechanism that is not illustrated in the drawing, and a tray **53**.

(39) The cleaning brush **51** is brought into contact with the medium transporting belt **10**, and cleans the adhesive layer F that is a front layer of the medium transporting belt **10**. The cleaning brush **51** rotates with the brush rotary shaft being the center. The brush rotary shaft may be a shaft or may be an imaginary center axis of rotation. The cleaning brush **51** is configured by connecting a plurality of cleaning rollers along the brush rotary shaft. The cleaning brush **51** cleans the adhesive layer F using a cleaning agent.

(40) The cleaning-brush driving mechanism causes the cleaning brush **51** to rotate. The cleaning-brush driving mechanism generates rotational force that causes the cleaning brush **51** to rotate. The cleaning-brush driving mechanism transmits the generated rotational force to the cleaning brush **51**. With the transmitted rotational force, the cleaning brush **51** rotates in a predetermined direction. The rotating cleaning brush **51** is brought into contact with the adhesive layer F of the medium transporting belt **10** to clean the adhesive layer F.

(41) The tray **53** stores a cleaning agent. The cleaning agent is a water-soluble solvent or the like such as water or alcohol aqueous solution. The cleaning agent may contain a surface-active agent, an anti-foaming agent, or the like. The tray **53** stores the cleaning agent supplied to the medium transporting belt **10** by the cleaning brush **51**.

(42) The cleaning unit **5** may include a blade or a height adjusting mechanism. The blade or the height adjusting mechanism is not illustrated in the drawing. The blade scrapes off dirt or the like attached on the cleaning brush **51**. The height adjusting mechanism adjusts the height of the cleaning brush **51** along the Z-axis. As the height adjusting mechanism adjusts the height of the cleaning brush **51** along the Z-axis, the state of contact of the cleaning brush **51** with the medium transporting belt **10** is adjusted.

(43) The winding unit **6** winds the printing medium M on which printing has performed at the printing unit **4**. The winding unit **6** includes a winding guide roller **61**, a winding shaft **63**, and a winding-shaft driving mechanism that is not illustrated in the drawing. The winding unit **6** may include a cutter **65**.

(44) The winding guide roller **61** is configured to guide the printing medium M on which printing has been performed at the printing unit **4**. The winding guide roller **61** guides the printing medium M transported in the transport direction A by the medium transporting belt **10**. The winding guide roller **61** guides the printing medium M toward the winding shaft **63** disposed at the $-Z$ direction relative to the winding guide roller **61**. The winding guide roller **61** may adjust the tension applied to the printing medium M. The winding guide roller **61** moves in the $+Z$ direction or the $-Z$ direction with a winding-tension adjusting mechanism that is not illustrated in the drawing. By moving in the $+Z$ direction or the $-Z$ direction, the winding guide roller **61** adjusts the tension applied to the printing medium M moving toward the winding shaft **63**.

(45) The winding shaft **63** winds the printing medium M on which printing has been performed at the printing unit **4**. By winding the printing medium M, the winding shaft **63** causes a printed-matter roll R2 to be formed. The winding shaft **63** rotatably supports the printed-matter roll R2. As

for the winding shaft **63**, a paper tube or the like may be installed by a user. The winding shaft **63** causes the printing medium **M** to be wound around the installed paper tube or the like. By rotating in a winding direction **D**, the winding shaft **63** winds the printing medium **M**.

(46) The winding-shaft driving mechanism causes the winding shaft **63** to rotate in the winding direction **D**. By causing the winding shaft **63** to rotate, the winding-shaft driving mechanism causes the winding shaft **63** to wind the printing medium **M** transported from the winding guide roller **61**. The winding-shaft driving mechanism may adjust the tension applied to the printing medium **M** that is transported by the winding unit **6**.

(47) The cutter **65** is configured to cut the printing medium **M** on which printing has been performed at the printing unit **4**. The cutter **65** is disposed downstream of the winding guide roller **61** in the transport direction **A**. The cutter **65** partially cuts the printing medium **M** in a direction along the **Y**-axis. The cutter **65** may cut the printing medium **M** along the transport direction **A**. The cutter **65** may not be provided.

First Embodiment

(48) The first embodiment describes a configuration in which one pressing unit **16** is provided on an inner side of the medium transporting belt **10**. FIGS. **2** and **3** illustrate the schematic configuration of the transport unit **3**. FIG. **2** illustrates the schematic configuration of the transport unit **3** as viewed from the **+Z** direction. FIG. **3** illustrates the schematic configuration of the transport unit **3** as viewed from the **+Y** direction. In FIG. **2**, neither the pressure applying roller **12** nor the upper surface portion of the medium transporting belt **10** is illustrated. The upper surface portion of the medium transporting belt **10** is a portion of the medium transporting belt **10** that is configured to transport the printing medium **M**. In FIG. **3**, the pressure applying roller **12** is not illustrated.

(49) FIG. **2** illustrates one pressing unit **16** configured to press the driving-roller circumferential surface **8a** of the driving roller **8**. FIG. **2** illustrates the driving roller **8**, the driven roller **9**, the lower surface portion of the medium transporting belt **10**, the support frame **14**, and the pressing unit **16**. The driving roller **8** corresponds to one example of a first roller. The driven roller **9** corresponds to one example of a second roller.

(50) The driving roller **8** includes the driving-roller circumferential surface **8a** and a driving roller shaft **8b**. The driving-roller circumferential surface **8a** corresponds to one example of a first circumferential surface. The medium transporting belt **10** is wound around the driving-roller circumferential surface **8a**. The rotational driving force from the driving mechanism is transmitted to the driving roller shaft **8b**. The driving roller shaft **8b** is rotatably supported by a main body frame or the like that is not illustrated in the drawing. The driving roller **8** rotates with the transmitted rotational driving force. With the rotation of the driving roller **8**, the medium transporting belt **10** rotationally moves.

(51) The driven roller **9** includes the driven-roller circumferential surface **9a** and a driven roller shaft **9b**. The driven-roller circumferential surface **9a** corresponds to one example of a second circumferential surface. The medium transporting belt **10** is wound around the driven-roller circumferential surface **9a**. The driven roller shaft **9b** is rotatably supported by the main body frame or the like. When the medium transporting belt **10** rotationally moves, the driven roller **9** followingly rotates.

(52) The driving roller **8** and the driven roller **9** give tension to the medium transporting belt **10**. The tension is given to the medium transporting belt **10** using a tension applying mechanism that is provided at the driving roller **8** or the driven roller **9** and is not illustrated in the drawing. With the tension given to the medium transporting belt **10**, deformation such as bending occurs in at least either of the driving roller **8** or the driven roller **9**. As one example, with the tension of the medium transporting belt **10**, the center portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** deforms in the **-X** direction. The central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** deforms in the **-X** direction, and bending occurs in the

driving roller shaft **8b** so as to have an arc shape.

(53) The support frame **14** supports the pressing unit **16**. The support frame **14** is disposed on an inner side of the medium transporting belt **10** wound around the driving roller **8** and the driven roller **9**. The support frame **14** is disposed between the driving roller **8** and the driven roller **9**.

(54) The pressing unit **16** presses a portion of the driving roller **8**. The pressing unit **16** is disposed on an inner side of the medium transporting belt **10**. The pressing unit **16** presses the driving-roller circumferential surface **8a** of the driving roller **8** from the inner side of the medium transporting belt **10**. The pressing unit **16** includes a bearing **17** and a bearing supporting member **19**. The bearing **17** is configured so as to include a driven portion **18**. The bearing **17** corresponds to one example of a bearing unit. The driven portion **18** corresponds to one example of a contact portion.

(55) The bearing **17** presses a portion of the driving roller **8**. The inner peripheral portion of the bearing **17** is supported by the bearing supporting member **19**. The driven portion **18** that constitutes the outer peripheral portion of the bearing **17** is rotatably supported. The bearing **17** includes a ball bearing, a cylindrical roller bearing, a needle roller bearing, a plain bearing, or the like.

(56) The driven portion **18** is brought into contact with the driving-roller circumferential surface **8a**. The driven portion **18** is included in the bearing **17**, and constitutes a portion of the bearing **17**. The driven portion **18** presses the driving-roller circumferential surface **8a**. With the rotation of the driving roller **8**, the driven portion **18** followingly rotates. The driven portion **18** illustrated in FIG. 2 is a pair of tubular members configured such that the bearing supporting member **19** is interposed between them. The driven portion **18** illustrated in FIG. 2 presses a central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b**.

(57) The bearing supporting member **19** supports the bearing **17**. The bearing **17** is supported at the end portion, in the +X direction, of the bearing supporting member **19**. The bearing supporting member **19** illustrated in FIG. 2 supports the bearing **17** at the central portion of the driven portion **18**. The end portion, in the -X direction, of the bearing supporting member **19** is supported by the support frame **14**. The length of the bearing supporting member **19** along the X-axis is adjusted on an as-necessary basis. By adjusting the length of the bearing supporting member **19**, a user may adjust the pressing force that the pressing unit **16** applies to the driving-roller circumferential surface **8a**.

(58) FIG. 3 illustrates the driving roller **8**, the driven roller **9**, the medium transporting belt **10**, the support frame **14**, and the pressing unit **16**. The support frame **14** and the pressing unit **16** are disposed on an inner side of the medium transporting belt **10**. The pressing unit **16** presses the driving-roller circumferential surface **8a** from the inner side of the medium transporting belt **10**. The pressing unit **16** presses the driving-roller circumferential surface **8a** on an imaginary horizontal line VH. The imaginary horizontal line VH is an imaginary line parallel to the X-axis passing through the center of rotation of the driving roller **8**.

(59) In a case of the driving roller **8**, the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** may deform in the -X direction due to the tension of the medium transporting belt **10**. When the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** deforms in the -X direction, the peripheral speed differs between the central portion and the end portion of the medium transporting belt **10** in the Y-axis direction. When the peripheral speed differs between the central portion and the end portion of the medium transporting belt **10** in the Y-axis direction, a difference occurs between an ink landing position at the central portion of the medium transporting belt **10** in the Y-axis direction and an ink landing position at the end portion of the medium transporting belt **10** in the Y-axis direction. The difference in the ink landing positions leads to a deterioration in the quality of the printed image. By causing the pressing unit **16** to press the position that deforms in the -X direction, a user causes the driving roller **8** to deform in the +X direction. By using the pressing unit **16**, the user is able to correct the deformation of the driving roller **8**. As the deformation of the driving roller **8** is corrected, the

difference in peripheral speeds in the medium transporting belt **10** occurring due to the deformation of the driving roller **8** reduces. As the difference in peripheral speeds reduces, a difference in ink landing positions within the medium transporting belt **10** reduces. This reduces a deterioration in quality occurring due to the difference in ink landing positions.

(60) The transport unit **3** includes: the medium transporting belt **10** having an endless shape and configured to transport the printing medium **M**; the driving roller **8** having the driving-roller circumferential surface **8a** brought into contact with the medium transporting belt **10**, the driving roller **8** being a roller around which the medium transporting belt **10** is wound; the driven roller **9** having the driven-roller circumferential surface **9a** brought into contact with the medium transporting belt **10**, the driven roller **9** being a roller around which the medium transporting belt **10** is wound; the pressing unit **16** including the bearing **17** including the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**, the pressing unit **16** being configured to press the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**; and the support frame **14** disposed on an inner side of the medium transporting belt **10** and configured to support the pressing unit **16**.

(61) The transport unit **3** is able to reduce the deformation of the driving roller **8** occurring at the time of extending the medium transporting belt **10** in a tension manner. The transport unit **3** is able to reduce the difference in peripheral speeds within the medium transporting belt **10**.

(62) The printer **1** includes: the medium transporting belt **10** having an endless shape and configured to transport the printing medium **M**; the driving roller **8** including the driving-roller circumferential surface **8a** brought into contact with the medium transporting belt **10**, the driving roller **8** being a roller around which the medium transporting belt **10** is wound; the driven roller **9** including the driven-roller circumferential surface **9a** brought into contact with the medium transporting belt **10**, the driven roller **9** being a roller around which the medium transporting belt **10** is wound; the pressing unit **16** including the bearing **17** including the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**, the pressing unit **16** being configured to press the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**; the support frame **14** disposed on an inner side of the medium transporting belt **10** and configured to support the pressing unit **16**; and the printing unit **4** configured to perform printing on the printing medium **M** transported by the medium transporting belt **10**.

(63) The printer **1** is able to reduce the deformation of the driving roller **8** occurring at the time of extending the medium transporting belt **10** in a tension manner. By reducing the deformation of the driving roller **8**, the printer **1** is able to reduce a deterioration in the quality of image.

(64) In the first embodiment, the pressing unit **16** presses the driving roller **8**. However, the configuration thereof is not limited to this. It may be possible to employ a configuration in which the pressing unit **16** presses the driven roller **9**. By pressing the driven roller **9**, the pressing unit **16** is able to reduce the deformation of the driven roller **9**. In a case of the configuration in which the pressing unit **16** presses the driven roller **9**, the driven roller **9** corresponds to one example of the first roller. The driven-roller circumferential surface **9a** that is brought into contact with the driven portion **18** of the pressing unit **16** corresponds to one example of the first circumferential surface. The driving roller **8** corresponds to one example of the second roller. The driving-roller circumferential surface **8a** corresponds to one example of the second circumferential surface.

(65) When the medium transporting belt **10** is extended in a tension manner by three or more rollers including the driving roller **8** and the driven roller **9**, the pressing unit **16** may press a roller differing from the driving roller **8** or the driven roller **9** as one example of the first roller.

Second Embodiment

(66) The second embodiment describes a configuration in which a plurality of pressing units **16** are disposed at the driving roller **8**. FIGS. **4** and **5** each illustrate the configuration example of the second embodiment. FIG. **4** illustrates a schematic configuration of the transport unit **3** as viewed from the +Y direction. FIG. **4** illustrates one example of the second embodiment. In FIG. **4**, the

pressure applying roller **12** is not illustrated. FIG. 5 illustrates a schematic configuration of the transport unit **3** as viewed from the +Z direction. FIG. 5 illustrates one example of the second embodiment. In FIG. 5, the pressure applying roller **12** or the upper surface portion of the medium transporting belt **10** is not illustrated.

(67) FIG. 4 illustrates the driving roller **8**, the driven roller **9**, the medium transporting belt **10**, the support frame **14**, and two pressing units **16**. The two pressing unit **16** are a first pressing unit **16a** and a second pressing unit **16b**. The driving roller **8**, the driven roller **9**, the medium transporting belt **10**, and the support frame **14** each have the same configuration as that in the first embodiment.

(68) The first pressing unit **16a** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The first pressing unit **16a** presses the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** as viewed from the +Z direction. The first pressing unit **16a** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The first pressing unit **16a** includes a first bearing **17a** and a first bearing supporting member **19a**. The configuration of the first pressing unit **16a** is the same as the configuration of the pressing unit **16** described in the first embodiment. The first pressing unit **16a** corresponds to one example of a pressing section. The first bearing **17a** corresponds to one example of a bearing unit.

(69) The first bearing **17a** includes a first driven portion **18a**. The first bearing **17a** is supported by the first bearing supporting member **19a**. The first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The first driven portion **18a** illustrated in FIG. 4 is brought into contact with the driving-roller circumferential surface **8a** at a position in the +Z direction relative to the imaginary horizontal line VH. The first driven portion **18a** corresponds to one example of a contact portion. The first bearing supporting member **19a** supports the first bearing **17a** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(70) The second pressing unit **16b** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The second pressing unit **16b** presses the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** as viewed from the +Z direction. The second pressing unit **16b** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The second pressing unit **16b** includes a second bearing **17b** and a second bearing supporting member **19b**. The configuration of the second pressing unit **16b** is the same as the configuration of the first pressing unit **16a**. The second pressing unit **16b** corresponds to one example of a second pressing section.

(71) The second bearing **17b** includes a second driven portion **18b**. The second bearing **17b** is supported by the second bearing supporting member **19b**. The second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a** at a position in the -Z direction relative to the imaginary horizontal line VH. The distance between a position where the second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a** and the imaginary horizontal line VH is equal to or substantially equal to the distance between a position where the first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a** and the imaginary horizontal line VH. The second bearing supporting member **19b** supports the second bearing **17b** at one end portion thereof, and is supported by the support frame **14** at the other end portion thereof. The second bearing **17b** corresponds to one example of a second bearing unit. The second driven portion **18b** corresponds to one example of a second contact portion.

(72) As illustrated in FIG. 4, the distance between a position where the second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a** and the imaginary horizontal line VH is equal to or substantially equal to the distance between a position where the first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a** and the

imaginary horizontal line VH. Thus, by adding the vector of force with which the second driven portion **18b** presses the driving-roller circumferential surface **8a** and the vector of force with which the first driven portion **18a** presses the driving-roller circumferential surface **8a**, the resultant vector is a vector of force with which the driving roller **8** is pressed in a direction equal to the imaginary horizontal line VH or in a direction substantially equal to the imaginary horizontal line VH. Thus, the first pressing unit **16a** and the second pressing unit **16b** are able to press the driving roller **8** in a stable state. As the first pressing unit **16a** and the second pressing unit **16b** press the driving roller **8**, the deformation of the driving roller **8** reduces. As the deformation of the driving roller **8** reduces, it is possible to reduce a deterioration in the quality of an image on the printed matter on which printing has been performed by the printer **1**.

(73) FIG. 5 illustrates the driving roller **8**, the driven roller **9**, the medium transporting belt **10**, the support frame **14**, and three pressing units **16**. The three pressing units **16** are a first pressing unit **16a**, a third pressing unit **16c**, and a fourth pressing unit **16d**. The driving roller **8**, the driven roller **9**, the medium transporting belt **10**, and the support frame **14** each have the same configuration as that in the first embodiment.

(74) The first pressing unit **16a** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The first pressing unit **16a** presses the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** as viewed from the +Z direction. The first pressing unit **16a** illustrated in FIG. 5 presses the driving roller **8** on the imaginary horizontal line VH as viewed from the +Y direction. The first pressing unit **16a** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The first pressing unit **16a** includes the first bearing **17a** and the first bearing supporting member **19a**. The configuration of the first pressing unit **16a** is the same as the configuration of the pressing unit **16** described in the first embodiment. The first pressing unit **16a** corresponds to one example of the pressing section. The first bearing **17a** corresponds to one example of the bearing unit.

(75) The first bearing **17a** includes the first driven portion **18a**. The first bearing **17a** is supported by the first bearing supporting member **19a**. The first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The first driven portion **18a** illustrated in FIG. 5 is brought into contact with the driving-roller circumferential surface **8a** at the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b**. The first driven portion **18a** illustrated in FIG. 5 is brought into contact with the driving-roller circumferential surface **8a** on the imaginary horizontal line VH as viewed from the +Y direction. The first driven portion **18a** corresponds to one example of the contact portion. The first bearing supporting member **19a** supports the first bearing **17a** at one end portion thereof, and is supported by the support frame **14** at the other end portion thereof.

(76) The third pressing unit **16c** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The third pressing unit **16c** presses a position of the driving roller **8** that is located more at the -Y direction than the central portion thereof in the axial direction of the driving roller shaft **8b**. The third pressing unit **16c** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The third pressing unit **16c** includes a third bearing **17c**, and a third bearing supporting member **19c**. The configuration of the third pressing unit **16c** is the same as the configuration of the first pressing unit **16a**. The third pressing unit **16c** corresponds to one example of the second pressing section.

(77) The third bearing **17c** includes a third driven portion **18c**. The third bearing **17c** is supported by the third bearing supporting member **19c**. The third driven portion **18c** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The third driven portion **18c** is brought into contact with the driving-roller circumferential surface **8a** at the -Y direction relative to the first driven portion **18a**. The third bearing supporting member **19c** supports the third bearing **17c** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction. The position, along the Y-

axis, where the third driven portion **18c** is brought into contact with the driving-roller circumferential surface **8a** is adjusted by a user on an as-necessary basis. The third bearing **17c** corresponds to one example of the second bearing unit. The third driven portion **18c** corresponds to one example of the second contact portion.

(78) The fourth pressing unit **16d** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The fourth pressing unit **16d** presses a position of the driving roller **8** that is located more at the +Y direction than the central portion thereof in the axial direction of the driving roller shaft **8b**. The fourth pressing unit **16d** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The fourth pressing unit **16d** includes a fourth bearing **17d** and a fourth bearing supporting member **19d**. The configuration of the fourth pressing unit **16d** is the same as the configuration of the first pressing unit **16a**.

(79) The fourth bearing **17d** includes a fourth driven portion **18d**. The fourth bearing **17d** is supported by the fourth bearing supporting member **19d**. The fourth driven portion **18d** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The fourth driven portion **18d** is brought into contact with the driving-roller circumferential surface **8a** at the +Y direction relative to the first driven portion **18a**. The fourth bearing supporting member **19d** supports the fourth bearing **17d** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction. The position, along the Y-axis, where the fourth driven portion **18d** is brought into contact with the driving-roller circumferential surface **8a** is adjusted by a user on an as-necessary basis.

(80) When the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** is deformed in the -X direction due to the tension of the medium transporting belt **10**, it is preferable that the first pressing unit **16a** is disposed at a position where the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** is pressed. It is preferable that the third pressing unit **16c** and the fourth pressing unit **16d** are disposed at a position of the -Y direction and a position of the +Y direction so as to be equidistantly spaced apart from the first pressing unit **16a** along the Y-axis.

(81) When the driving roller **8** is deformed in the -X direction at a plurality of positions in the axial direction of the driving roller shaft **8b** due to the tension of the medium transporting belt **10**, it is preferable that the first pressing unit **16a**, the third pressing unit **16c**, and the fourth pressing unit **16d** are disposed at positions, in the axial direction of the driving roller shaft **8b**, where the deformation in the -X direction occurs. The plurality of pressing units **16** are each disposed by a user as appropriate. In FIG. 5, three pressing units **16** are disposed. However, the configuration is not limited to this. Four or more pressing units **16** may be disposed.

(82) It is preferable that the transport unit **3** includes the third pressing unit **16c** including the third bearing **17c** including the third driven portion **18c** brought into contact with the driving-roller circumferential surface **8a**, the third pressing unit **16c** being configured to press the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**.

(83) By pressing the driving roller **8** using two pressing units **16**, the transport unit **3** is able to further reduce the deformation of the driving roller **8**. As the deformation of the driving roller **8** reduces, it is possible to reduce a difference in peripheral speeds within the medium transporting belt **10**.

(84) In FIGS. 4 and 5, the driving roller **8** is pressed by the plurality of pressing units **16**. However, the configuration is not limited to this. The plurality of pressing units **16** may press the driven roller **9**. By pressing the driven roller **9** using the plurality of pressing units **16**, it is possible to reduce the deformation of the driven roller **9**. In addition, the configurations illustrated in FIGS. 4 and 5 may be combined on an as-necessary basis.

Third Embodiment

(85) The third embodiment describes a configuration in which the pressing unit **16** is disposed at each of the driving roller **8** and the driven roller **9**. FIGS. 6 and 7 each illustrate the configuration

example of the third embodiment. FIG. 6 illustrates the schematic configuration of the transport unit 3 as viewed from the +Z direction. FIG. 6 illustrates one example of the third embodiment. In FIG. 6, neither the pressure applying roller 12 nor the upper surface portion of the medium transporting belt 10 is illustrated. FIG. 7 illustrates the schematic configuration of the transport unit 3 as viewed from the +Y direction. FIG. 7 illustrates one example of the third embodiment. In FIG. 7, the pressure applying roller 12 is not illustrated.

(86) FIG. 6 illustrates the driving roller 8, the driven roller 9, the medium transporting belt 10, the support frame 14, and two pressing units 16. The two pressing units 16 are the first pressing unit 16a, and a fifth pressing unit 16e. The driving roller 8, the driven roller 9, the medium transporting belt 10, and the support frame 14 each have the same configuration as that in the first embodiment.

(87) The first pressing unit 16a presses a portion of the driving-roller circumferential surface 8a of the driving roller 8. The first pressing unit 16a presses the central portion of the driving roller 8 in the axial direction of the driving roller shaft 8b. The first pressing unit 16a presses the driving-roller circumferential surface 8a on an inner side of the medium transporting belt 10. The first pressing unit 16a includes the first bearing 17a and the first bearing supporting member 19a. The configuration of the first pressing unit 16a is the same as the configuration of the pressing unit 16 described in the first embodiment. The first pressing unit 16a corresponds to one example of the pressing section. The first bearing 17a corresponds to one example of the bearing unit.

(88) The first bearing 17a includes the first driven portion 18a. The first bearing 17a is supported by the first bearing supporting member 19a. The first driven portion 18a is brought into contact with the driving-roller circumferential surface 8a, and presses the driving roller 8. The first driven portion 18a illustrated in FIG. 6 is brought into contact with the driving-roller circumferential surface 8a at the central portion of the driving roller 8 in the axial direction of the driving roller shaft 8b. The first driven portion 18a corresponds to one example of the contact portion. The first bearing supporting member 19a supports the first bearing 17a at one end portion thereof in the X-axis direction, and is supported by the support frame 14 at the other end portion thereof in the X-axis direction.

(89) The fifth pressing unit 16e presses a portion of the driven-roller circumferential surface 9a of the driven roller 9. The fifth pressing unit 16e presses the central portion of the driven roller 9 in the axial direction of the driving roller shaft 8b. The fifth pressing unit 16e presses the driven-roller circumferential surface 9a on an inner side of the medium transporting belt 10. The fifth pressing unit 16e includes a fifth bearing 17e and a fifth bearing supporting member 19e. The configuration of the fifth pressing unit 16e is the same as the configuration of the first pressing unit 16a. The fifth pressing unit 16e corresponds to one example of a second-roller pressing section. The fifth bearing 17e corresponds to one example of a second-roller bearing unit.

(90) The fifth bearing 17e includes a fifth driven portion 18e. The fifth bearing 17e is supported by the fifth bearing supporting member 19e. The fifth driven portion 18e corresponds to one example of a second-roller contact portion. The fifth driven portion 18e is brought into contact with the driven-roller circumferential surface 9a, and presses the driven roller 9. The fifth driven portion 18e is brought into contact with the driven-roller circumferential surface 9a at the central portion of the driven roller 9 in the axial direction of the driven roller shaft 9b. The fifth bearing supporting member 19e supports the fifth bearing 17e at one end portion thereof in the X-axis direction, and is supported by the support frame 14 at the other end portion thereof in the X-axis direction.

(91) The first pressing unit 16a reduces the deformation in the -X direction occurring in the driving roller 8 due to the tension of the medium transporting belt 10. The fifth pressing unit 16e reduces the deformation in the +X direction occurring in the driven roller 9 due to the tension of the medium transporting belt 10. As the first pressing unit 16a and the fifth pressing unit 16e are provided, it is possible to reduce a difference in peripheral speeds occurring in the medium transporting belt 10 between the belt central portion and the belt end portion of the medium transporting belt 10.

(92) The transport unit **3** includes the fifth pressing unit **16e** including the fifth bearing **17e** including the fifth driven portion **18e** brought into contact with the driven-roller circumferential surface **9a**, the fifth pressing unit **16e** being configured to press the driven-roller circumferential surface **9a** on an inner side of the medium transporting belt **10**.

(93) As the fifth pressing unit **16e** is disposed relative to the driven roller **9**, it is possible to reduce the deformation of the driven roller **9**.

(94) FIG. 7 illustrates the driving roller **8**, the driven roller **9**, the medium transporting belt **10**, the support frame **14**, and four pressing units **16**. The four pressing units **16** are the first pressing unit **16a**, the second pressing unit **16b**, the fifth pressing unit **16e**, and a sixth pressing unit **16f**. The driving roller **8**, the driven roller **9**, the medium transporting belt **10**, and the support frame **14** each have the same configuration as that in the first embodiment.

(95) The first pressing unit **16a** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The first pressing unit **16a** presses the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** as viewed from the +Z direction. The first pressing unit **16a** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The first pressing unit **16a** includes the first bearing **17a** and the first bearing supporting member **19a**. The first bearing **17a** illustrated in FIG. 7 is supported by the first bearing supporting member **19a**. The end portion, in the +Y direction, of the first bearing **17a** and the end portion thereof in the -Y direction are supported by the first bearing supporting member **19a**. The first pressing unit **16a** corresponds to one example of the pressing section. The first bearing **17a** corresponds to one example of the bearing unit.

(96) The first bearing **17a** includes the first driven portion **18a**. The first bearing **17a** is supported by the first bearing supporting member **19a**. The first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The first driven portion **18a** illustrated in FIG. 7 is brought into contact with the driving-roller circumferential surface **8a** at the +Z direction relative to the imaginary horizontal line VH. The first driven portion **18a** corresponds to one example of the contact portion. The first bearing supporting member **19a** supports the first bearing **17a** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(97) The second pressing unit **16b** presses a portion of the driving-roller circumferential surface **8a** of the driving roller **8**. The second pressing unit **16b** presses the central portion of the driving roller **8** in the axial direction of the driving roller shaft **8b** as viewed from the +Z direction. The second pressing unit **16b** presses the driving-roller circumferential surface **8a** on an inner side of the medium transporting belt **10**. The second pressing unit **16b** includes the second bearing **17b** and the second bearing supporting member **19b**. The second bearing **17b** illustrated in FIG. 7 is supported by the second bearing supporting member **19b**. The end portion, at the +Y direction, of the second bearing **17b** and the end portion thereof at the -Y direction are supported by the second bearing supporting member **19b**. The configuration of the second pressing unit **16b** is the same as the configuration of the first pressing unit **16a** illustrated in FIG. 7.

(98) The second bearing **17b** includes the second driven portion **18b**. The second bearing **17b** is supported by the second bearing supporting member **19b**. The second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a**, and presses the driving roller **8**. The second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a** at the -Z direction relative to the imaginary horizontal line VH. The distance between a position where the second driven portion **18b** is brought into contact with the driving-roller circumferential surface **8a** and the imaginary horizontal line VH is equal to or substantially equal to the distance between a position where the first driven portion **18a** is brought into contact with the driving-roller circumferential surface **8a** and the imaginary horizontal line VH. The second bearing supporting member **19b** supports the second bearing **17b** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(99) The fifth pressing unit **16e** presses a portion of the driven-roller circumferential surface **9a** of the driven roller **9**. The fifth pressing unit **16e** presses the central portion of the driven roller **9** in the axial direction of the driven roller shaft **9b** as viewed from the +Z direction. The fifth pressing unit **16e** presses the driven-roller circumferential surface **9a** on an inner side of the medium transporting belt **10**. The fifth pressing unit **16e** includes the fifth bearing **17e** and the fifth bearing supporting member **19e**. The fifth bearing **17e** illustrated in FIG. 7 is supported by the fifth bearing supporting member **19e**. The end portion, in the +Y direction, of the fifth bearing **17e** and the end portion thereof in the -Y direction are supported by the fifth bearing supporting member **19e**. The configuration of the fifth pressing unit **16e** is the same as the configuration of the first pressing unit **16a** illustrated in FIG. 7. The fifth pressing unit **16e** corresponds to one example of the second-roller pressing section. The fifth bearing **17e** corresponds to one example of the second-roller bearing unit.

(100) The fifth bearing **17e** includes the fifth driven portion **18e**. The fifth bearing **17e** is supported by the fifth bearing supporting member **19e**. The fifth driven portion **18e** is brought into contact with the driven-roller circumferential surface **9a**, and presses the driven roller **9**. The fifth driven portion **18e** illustrated in FIG. 7 is brought into contact with the driven-roller circumferential surface **9a** at the +Z direction relative to the imaginary horizontal line VH. The fifth driven portion **18e** corresponds to one example of the second-roller contact portion. The fifth bearing supporting member **19e** supports the fifth bearing **17e** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(101) The sixth pressing unit **16f** presses a portion of the driven-roller circumferential surface **9a** of the driven roller **9**. The sixth pressing unit **16f** presses the central portion of the driven roller **9** in the axial direction of the driven roller shaft **9b** as viewed from the +Z direction. The sixth pressing unit **16f** presses the driven-roller circumferential surface **9a** on an inner side of the medium transporting belt **10**. The sixth pressing unit **16f** includes a sixth bearing **17f** and a sixth bearing supporting member **19f**. The sixth bearing **17f** illustrated in FIG. 7 is supported by the sixth bearing supporting member **19f**. The end portion, in the +Y direction, of the sixth bearing **17f** and the end portion thereof in the -Y direction are supported by the sixth bearing supporting member **19f**. The configuration of the sixth pressing unit **16f** is the same as the configuration of the fifth pressing unit **16e** illustrated in FIG. 7.

(102) The sixth bearing **17f** includes a sixth driven portion **18f**. The sixth bearing **17f** is supported by the sixth bearing supporting member **19f**. The sixth driven portion **18f** is brought into contact with the driven-roller circumferential surface **9a**, and presses the driven roller **9**. The sixth driven portion **18f** is brought into contact with the driven-roller circumferential surface **9a** at the -Z direction relative to the imaginary horizontal line VH. The distance between a position where the sixth driven portion **18f** is brought into contact with the driven-roller circumferential surface **9a** and the imaginary horizontal line VH is equal to or substantially equal to the distance between a position where the fifth driven portion **18e** is brought into contact with the driven-roller circumferential surface **9a** and the imaginary horizontal line VH. The sixth bearing supporting member **19f** supports the sixth bearing **17f** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

Fourth Embodiment

(103) A fourth embodiment describes the schematic configuration of the transport unit **3** including a pressing-force adjusting mechanism. FIGS. **8**, **9**, **10**, and **11** each illustrate the schematic configuration of the transport unit **3** including the pressing-force adjusting mechanism. The pressing-force adjusting mechanism corresponds to one example of an adjustment unit. FIGS. **8**, **9**, **10**, and **11** each illustrate the schematic configuration of the transport unit **3** as viewed from the +Y direction. In FIGS. **8**, **9**, **10**, and **11**, the pressure applying roller **12** is not illustrated.

(104) FIGS. **8**, **9**, **10**, and **11** each illustrate the driving roller **8**, the driven roller **9**, the medium transporting belt **10**, the support frame **14**, and the pressing unit **16**. The driving roller **8**, the driven

roller **9**, the medium transporting belt **10**, and the support frame **14** each have the same configuration as that in the first embodiment.

(105) FIG. **8** illustrates the schematic configuration of the transport unit **3** including the pressing-force adjusting mechanism. FIG. **8** illustrates the configuration in which the pressing unit **16** includes the pressing-force adjusting mechanism. FIG. **8** illustrates a screw mechanism **81** serving as the pressing-force adjusting mechanism.

(106) The pressing unit **16** illustrated in FIG. **8** includes two bearings **17**, the bearing supporting member **19**, and the screw mechanism **81**. As illustrated in FIG. **8**, the pressing unit **16** includes two or more bearings **17**. The two bearings **17** each include the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**. The two bearings **17** is supported by the bearing supporting member **19** through the screw mechanism **81**. The bearing supporting member **19** has one end portion configured to support the two bearings **17** through the screw mechanism **81**, and has the other end portion configured to be supported by the support frame **14**.

(107) The screw mechanism **81** is configured to adjust the pressing force with which the pressing unit **16** presses the driving roller **8**. The screw mechanism **81** is provided between the bearings **17** and the bearing supporting member **19**. The screw mechanism **81** includes a screw portion **81a** and a screw hole **81b**. One end portion of the screw portion **81a** in the X-axis direction is configured to support the bearings **17**. The other end portion of the screw portion **81a** in the X-axis direction is configured to have a spiral groove formed thereon. The other end portion of the screw portion **81a** in the X-axis direction is fitted into the screw hole **81b**. As a user adjusts the amount of fitting of the screw portion **81a** fitted into the screw hole **81b**, the pressing force of the driven portion **18** that presses against the driving-roller circumferential surface **8a** is adjusted.

(108) FIG. **9** illustrates the schematic configuration of the transport unit **3** including the pressing-force adjusting mechanism. FIG. **9** illustrates a configuration in which the pressing unit **16** includes the pressing-force adjusting mechanism. FIG. **9** illustrates an air cylinder **83** serving as the pressing-force adjusting mechanism.

(109) The pressing unit **16** illustrated in FIG. **9** includes two bearings **17**, the bearing supporting member **19**, and the air cylinder **83**. The two bearings **17** each include the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**. The two bearings **17** are supported by the bearing supporting member **19** through the air cylinder **83**. The bearing supporting member **19** supports, through the air cylinder **83**, the two bearings **17** at one end portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(110) The air cylinder **83** is configured to adjust the pressing force with which the driving roller **8** is pressed by the pressing unit **16**. One end portion of the air cylinder **83** in the X-axis direction supports the bearing **17**. The other end portion of the air cylinder **83** in the X-axis direction is supported by the bearing supporting member **19**. The air pressure within the air cylinder **83** is adjusted to adjust the pressing force with which the driven portion **18** presses the driving-roller circumferential surface **8a**. The air pressure within the air cylinder **83** may be adjusted by manual operation of a user. The air pressure within the air cylinder **83** may be adjusted so as to be a desired air pressure by using a control mechanism that is not illustrated in the drawing.

(111) FIG. **10** illustrates the schematic configuration of the transport unit **3** including the pressing-force adjusting mechanism. FIG. **10** illustrates a configuration in which the pressing unit **16** includes the pressing-force adjusting mechanism. FIG. **10** illustrates a cam mechanism **85** serving as the pressing-force adjusting mechanism.

(112) The pressing unit **16** illustrated in FIG. **10** includes two bearings **17**, the bearing supporting member **19**, and the cam mechanism **85**. The two bearings **17** each include the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**. The two bearings **17** are supported by the bearing supporting member **19** through the cam mechanism **85**. The bearing supporting member **19** supports the two bearings **17** through the cam mechanism **85** at one end

portion thereof in the X-axis direction, and is supported by the support frame **14** at the other end portion thereof in the X-axis direction.

(113) The cam mechanism **85** is configured to adjust the pressing force with which the driving roller **8** is pressed by the pressing unit **16**. The cam mechanism **85** is supported by the bearing supporting member **19**. The cam mechanism **85** includes a cam member **85a** and a cam rotary shaft **85b**. With driving of a cam-rotation driving mechanism that is not illustrated in the drawing, the cam member **85a** rotates. The cam rotary shaft **85b** supports the cam member **85a**. The cam rotary shaft **85b** is a rotary shaft of the cam member **85a**. The cam member **85a** rotates with the cam rotary shaft **85b** being the center. As one example, the cam mechanism **85** is configured such that the angle of rotation of the cam member **85a** is adjusted by a mechanism for controlling the angle of rotation, which is not illustrated in the drawing. By adjusting the angle of rotation of the cam member **85a**, the pressing force with which the driven portion **18** presses against the driving-roller circumferential surface **8a** is adjusted.

(114) FIG. **11** illustrates the schematic configuration of the transport unit **3** including the pressing-force adjusting mechanism. FIG. **11** illustrates a configuration in which the pressing unit **16** includes the pressing-force adjusting mechanism. FIG. **11** illustrates an electrically powered actuator **87** serving as the pressing-force adjusting mechanism.

(115) The pressing unit **16** illustrated in FIG. **11** includes two bearings **17** and the electrically powered actuator **87**. The two bearings **17** each include the driven portion **18** brought into contact with the driving-roller circumferential surface **8a**. The two bearings **17** are supported by the electrically powered actuator **87**. The electrically powered actuator **87** is supported by the support frame **14**.

(116) The electrically powered actuator **87** is configured to adjust the pressing force with which the driving roller **8** is pressed by the pressing unit **16**. The electrically powered actuator **87** includes a rod **87a**, a ball screw **87b**, and a motor **87c**. The rod **87a** supports the two bearings **17**. With rotation of the ball screw **87b**, the rod **87a** moves in the +X direction or the -X direction. The ball screw **87b** rotates with rotational force generated at the motor **87c**. As the ball screw **87b** rotates to move the rod **87a** along the X-axis. The motor **87c** generates the rotational force that causes the ball screw **87b** to rotate. With a motor controlling mechanism that is not illustrated in the drawing, the motor rotational direction and the motor rotational speed of the motor **87c** are controlled. With the motor controlling mechanism, the position of the electrically powered actuator **87** is adjusted along the X-axis of the rod **87a**. As the position of the rod **87a** along the X-axis is adjusted, the pressing force with which the driven portion **18** presses against the driving-roller circumferential surface **8a** is adjusted.

(117) The transport unit **3** includes the pressing-force adjusting mechanism provided at the pressing unit **16** and configured to adjust the pressing force with the pressing unit **16**.

(118) As the pressing-force adjusting mechanism adjusts the pressing force with which the driving roller **8** is pressed, it is possible to adjust the amount of deformation of the driving roller **8** occurring due to the medium transporting belt **10**.

(119) FIGS. **8**, **9**, **10**, and **11** each illustrate a configuration in which the pressing unit **16** including the pressing-force adjusting mechanism presses the driving roller **8**. However, the configuration is not limited to this. The pressing unit **16** including the pressing-force adjusting mechanism may press the driven roller **9**. The pressing-force adjusting mechanism illustrated in FIGS. **8**, **9**, **10**, and **11** may be used in the pressing unit **16** described in the first embodiment, the second embodiment, or the third embodiment.

Claims

1. A transport device comprising: a transporting belt having an endless shape and configured to transport a medium; a first roller including a first circumferential surface brought into contact with

the transporting belt, the first roller being a roller around which the transporting belt is wound; a second roller including a second circumferential surface brought into contact with the transporting belt, the second roller being a roller around which the transporting belt is wound; a pressing section including a bearing unit including a contact portion brought into contact with the first circumferential surface, the pressing section being configured to press the first circumferential surface on an inner side of the transporting belt; and a support portion disposed on an inner side of the transporting belt and configured to support the pressing section.

2. The transport device according to claim 1, comprising an adjustment unit provided at the pressing unit and configured to adjust a pressing force by the pressing unit.

3. The transport device according to claim 1, comprising a second pressing section including a second bearing unit including a second contact portion brought into contact with the first circumferential surface, the second pressing section being configured to press the first circumferential surface on an inner side of the transporting belt.

4. The transport device according to claim 1, comprising a second-roller pressing section including a second-roller bearing unit including a second-roller contact portion brought into contact with the second circumferential surface, the second-roller pressing section being configured to press the second circumferential surface on an inner side of the transporting belt.

5. The transport device according to claim 1, wherein the first roller is a crown roller.

6. A recording device comprising: a transporting belt having an endless shape and configured to transport a medium; a first roller including a first circumferential surface brought into contact with the transporting belt, the first roller being a roller around which the transporting belt is wound; a second roller including a second circumferential surface brought into contact with the transporting belt, the second roller being a roller around which the transporting belt is wound; a pressing section including a bearing unit including a contact portion brought into contact with the first circumferential surface, the pressing section being configured to press the first circumferential surface on an inner side of the transporting belt; a support portion disposed on an inner side of the transporting belt and configured to support the pressing section; and a recording unit configured to perform recording on the medium transported by the transporting belt.
